LESSON 1
PROJECT MANAGEMENT

INTRODUCTION

A project is composed of jobs, activities, functions or tasks that are related one to the other in some manner, and all of these should be completed in order to complete the project. Every project has one specific purpose: it starts at some specific moment and it is finished when the objectives have been fulfilled.

PROJECT MANAGEMENT

Project management is the practice of initiating, planning, executing, controlling, and closing the work of a team to achieve specific goals and meet specific success criteria at the specified time. The primary challenge of project management is to achieve all of the project goals within the given constraints.

For completion of a project, two basic things are required:

i) Material resources,
ii) Manpower resources.

OBJECTIVES

Each project, whether big or small has three objectives:
i) The project should be completed be with a minimum of elapsed time. ii) It should use available manpower and other resources as sparingly as possible, without delay.
iii) It should be completed with a minimum of capital investment, without delay.

PHASES OF PROJECT MANAGEMENT

Project management involves the following three phases
i) Project planning
ii) Project schedule
iii) Project controlling.

Project Planning:

Planning is the most important phase of project management. Planning involves defining objectives of the project, listing of tasks or jobs that must be performed, determining gross requirements for material, equipment and manpower and preparing estimates of costs and durations for the various jobs or activities to bring about the satisfactory completion of the project.

Planning is important because :

(*i*) It provides direction

(*ii*) It provides unifying frame-work

(*iii*) It helps to reveal future opportunities and threats

(*iv*) It provides performance standards.

Scheduling:

Scheduling is the allocation of resources. These resources, in conceptual sense, are time and energy, but in practical sense are *time, space, equipment* and *effort* applied to material. More specifically, *scheduling* is the mechanical process of formalising the planned functions, assigning the starting and completion dates to each part (or activity) of the work in such a manner that the whole work (or project) proceeds in a logical sequence and in an orderly and semmetric manner.

Controlling:

Controlling consists of reviewing the difference between the schedule and actual performance once the project has begun. *Project control is the formal mechanism established to determine deviations from the basic plan, to determine the precise effect of these deviations on the plan, and to replan and reschedule to compensate for the deviations.*

PROJECT LIFE CYCLE

The project life cycle may be viewed as a process through which a project is implemented from beginning to end. This process is often very complex; however, it can be decomposed into several stages/phases.

- Preconstruction stage
- Procurement stage (Bidding and award stage)
- Construction stage

- Closeout stage

The solutions at various stages are then integrated to obtain the final outcome. Although each stage requires different expertise, it usually includes both technical and managerial activities in the knowledge domain of the specialist.

Preconstruction stage

The preconstruction phase of a project can be broken into conceptual planning, schematic design, design development, and contract documents.

Conceptual design:

- Very important for the owner.
- During this stage the owner hires key consultants including the designer and project manager, selects the project site, and establish a conceptual estimate, schedule, and program.
- The owner must gather as much information as possible about the project. - The most important decision is to proceed with the project or not.

Schematic design:

- During this stage, the project team investigates alternate design solutions, materials and systems.
- Completion of this stage represents about 30% of the design completion for the project.
- Designing the main systems and components of the project.
- Good communication between owner, designer, and construction manager is critical during this stage because selections during this design stage affect project appearance, construction and cost.
- This stage takes the project from 30% design to 60% design.

Contract documents:

- Final preparation of the documents necessary for the bid package such as the drawings, specifications, general conditions, and bill of quantities.
- All documents need to be closely reviewed by the construction manager and appropriate owner personnel to decrease conflicts, and changes.
- With the contract documents are almost complete; a detailed and complete cost estimate for the project can be done.

Procurement Stage (Bidding and award stage)

- The project formally transits from design into construction.
- This stage begins with a public advertisement for all interested bidders or an invitation for specific bidders.
- In fast-track projects, this stage overlaps with the design stage.
- If the project is phased, each work package will be advertised and bid out individually. - It is very important stage to select highly qualified contractors. It is not wise to select the under-bid contractors.

Construction stage

- The actual physical construction of the project stage.
- This stage takes the project from procurement through the final completion.
- It is the time where the bulk of the owner's funds will be spent.
- It is the outcome of all previous stages (i.e., good preparation means smooth construction).
- The consultant will be deployed for contract administration and construction supervision. - Changes during construction may hinder the progress of the project.

Closeout stage

- Transition from design and construction to the actual use of the constructed facility. - In this stage, the management team must provide documentation, shop drawings, as-built drawings, and operation manuals to the owner organization.
- The as-built drawings are the original contract drawings adjusted to reflect all the changes that occurred.
- Assessment of the project team's performance is crucial in this stage for avoiding mistakes in the future.
- Actual activity costs and durations should be recorded and compared with that was planned. This updated costs and durations will serve as the basis for the estimating and scheduling of future projects.

THE DESIGN PROFESSIONALS

The Project Manager: The project manager is the individual charged with the overall

coordination of the entire construction program for the owner. These include planning, design, procurement, and construction. Among his/her duties:

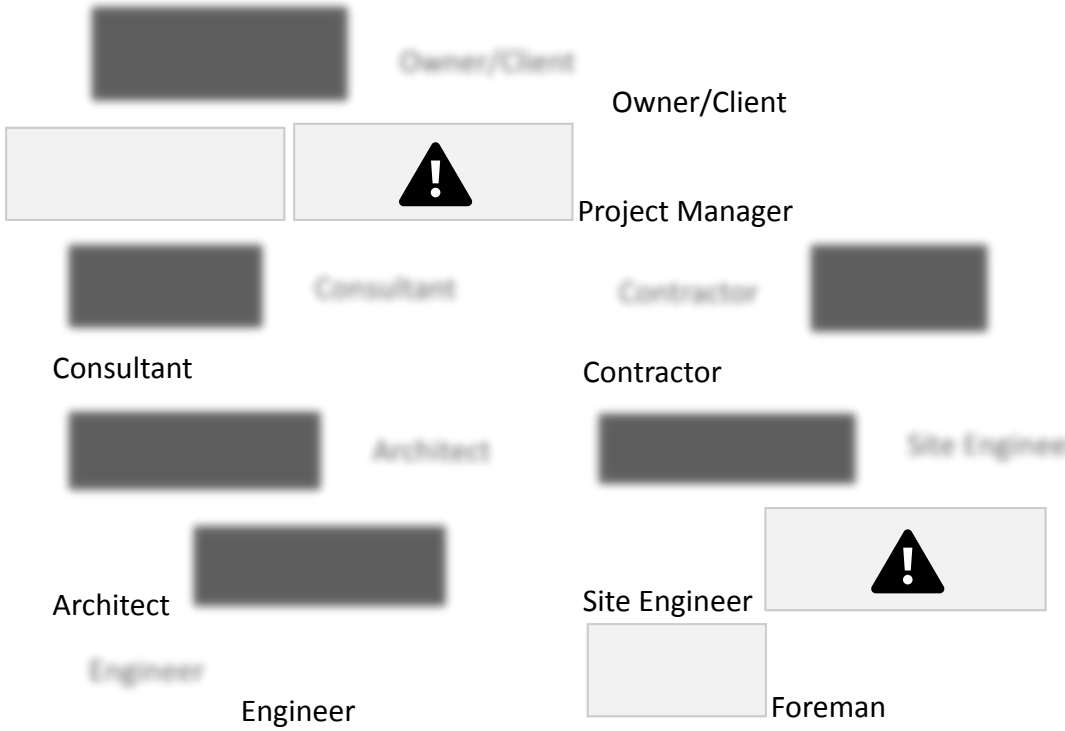
- Clear definitions of the goals of the project.

- Investigate alternative solutions for the problems.
- Develop a detailed plan to make the selected program reality.
- Implement the plan and control the project.

Architect: An architect is an individual who plans and design buildings and their associated landscaping. Architects mostly rely on consulting engineers for structural, electrical, and mechanical work.

Engineer: The term engineer usually refers to an individual or a firm engaged in the design or other work associated with the design or construction. Design engineers are usually classified as civil, electrical, mechanical depending upon their specialty. There are also scheduling, estimating, cost, and construction engineers.

Engineering-Construction Firm: An engineering-construction firm is a type of organization the combines both architect/engineering and construction contracting. This type of company has the ability of executing a complete design-build sequence.





Workers

Figure: Project Team Hierarchy (Small to Medium project)

CONFLICT & CONFLICT MANAGEMENT

Conflict is an inevitable and unavoidable part of our everyday professional and personal lives.

Conflict management is the process of limiting the negative aspects of conflict while increasing the positive aspects of conflict.

The aim of conflict management is to enhance learning and group outcomes, including effectiveness or performance in an organizational setting.

CAUSES OF CONFLICT

- Misunderstanding
- Personality clashes
- Competition for resources
- Authority issues
- Lack of cooperation
- Differences over methods or style
- Low performance
- Value or goal differences

Assumptions about, Causes of and Value of Conflict

	nflict What Does It Mean?	What is the Value to You?
	ding When individuals do not hear what is being said?.	
Personality Clashes	When individuals do not value “people just like me”.	
Competition for resources	When employees believe they are better off competing for resources rather than cooperating.	

Authority Issues	When employees lack confidence in their leaders or perceive overuse of authority.	
Lack of cooperation	When one person does not share information with the whole group.	
Differences over methods or style	When agreement does not exist on standard ways of completing a task.	
	ınce When individuals are not working to their	
Value or goal differences	When individuals value different outcomes or objectives.	

CONFLICT: CONSTRUCTIVE VS DESTRUCTIVE

Conflict is destructive when it:

— Diverts energy from more important issues and tasks.

— Deepens differences in values.

— Polarizes groups so that cooperation is reduced.

— Destroys the morale of people or reinforces poor self- concepts.

Conflict is constructive when it:

— Opens up issues of importance, resulting in issue clarification. — Helps build cohesiveness as people learn more about each other. — Causes reassessment by allowing for examination of procedures or actions. — Increases individual involvement.

CONFLICT MANAGEMENT STYLES

— Forcing conflict style: user attempts to resolve conflict by using aggressive behavior.

— Avoiding conflict style: user attempts to passively ignore the conflict rather than resolve it.

— Accommodating conflict style: user attempts to resolve the conflict by passively giving in to the other party.

— Compromising conflict style: user attempts to resolve the conflict through assertive give-and-take concessions.

— Collaborating conflict style: user assertively attempts to jointly resolve the

conflict with the best solution agreeable to all parties.

SOME TIPS FOR MINIMISING WORKPLACE CONFLICT

— Build good relationships before conflict occurs

— Do not let small problems escalate; deal with them as they arise

— Respect differences

— Listen to others' perspectives on the conflict situation

— Acknowledge feelings before focussing on facts

— Focus on solving problems, not changing people

— If you can't resolve the problem, turn to someone who can help —

Remember to adapt your style to the situation and persons involved

Questions:

1. What is project management and write the objectives of project management?
2. Describe the phases of project management.
3. What do you mean by project planning and project schedule?

   Or, Write short note on project planning and project schedule.
4. What is mean project controlling?

   Or, Write short note on project controlling.
5. Describe briefly/shortly the stages of project life cycle.
6. Describe the stages of preconstruction stage/phase of project life cycle.
7. What are the roles of Project Manager, Engineer, and Architect? 8.

Draw the flow chart project team Hierarchy.

9. What is conflict management? Why conflict occurs?
10. When Conflict can be constructive and destructive?
11. What are the styles for managing workplace conflict?
12. What are the tips for minimizing workplace conflict?